

Omid Navadehrazi

AI Engineer | LLM & RAG Systems | Computer Vision & Edge AI

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LinkedIn | GitHub | Portfolio

Authorized to work in Italy without employer sponsorship | Open to opportunities across Europe

Profile

AI Engineer building and deploying LLM, RAG, Computer Vision, Edge AI, and distributed ML systems. Current work focuses on local model serving, retrieval pipelines, streaming APIs, conversation state, GPU inference, benchmarking, and end-to-end integration for latency-sensitive AI applications. Experience spans model evaluation, system architecture, deployment, profiling, validation, and cross-team integration across cloud, GPU, embedded, and distributed environments.

Professional Experience

Artificial Intelligence Engineer

Jul 2026 – Present

Kalliope | University of Pisa – Industrial Collaboration

Pisa, Italy

- Engineered and integrated a local LLM service for a real-time AI platform using **Qwen2.5-7B**, **Ollama**, **FastAPI**, **Docker**, **FAISS**, and **OpenAI-compatible streaming APIs**.
- Designed modular RAG and conversation-state components using **multilingual embeddings** and **FAISS**, with relevance filtering, persistent knowledge management, isolated session memory, persistence, and cleanup.
- Validated end-to-end integration on **AWS with an NVIDIA A10G** across **2/2 successful integration calls and 16 conversation turns**; streaming, RAG, memory, session isolation, and cleanup all passed.
- Measured approximately **375 ms warm mean TTFT**, **398 ms warm P95 TTFT**, **890 ms mean generation time**, and **1.19 s P95 generation time** in the verified AWS integration without model fine-tuning.
- Benchmarked concurrency from **1–8 simultaneous requests** with **100% successful requests, zero timeouts**, and approximately **6.7 requests/s** observed throughput; also evaluated local LLM candidates for Italian conversational quality across CPU, RTX 5070, lab GPU, and AWS environments.

Artificial Intelligence Engineer

Mar 2025 – Jun 2026

CNIT – Consorzio Nazionale Interuniversitario per le Telecomunicazioni

Pisa, Italy

- Developed Computer Vision pipelines for **object detection**, **semantic segmentation**, and **behavior analysis** using **PyTorch**, **YOLO**, **OpenCV**, and **Roboflow** for intelligent monitoring applications.
- Built a reproducible Edge AI deployment workflow from **PyTorch/YOLO to ONNX**, **calibration**, **INT8 quantization**, **HAR/HEF compilation**, **Hailo emulation**, **Hailo-15H inference**, and **GStreamer** video pipelines.
- Co-developed a custom standalone Edge AI prototype from early hardware integration through field testing, contributing to hardware assembly, camera validation, system integration, and weekly technical feedback with the hardware partner.
- Field-tested **4 physical prototypes across 2 outdoor locations in Italy**, validating standalone camera acquisition, local AI processing, connectivity, and system operation under real-world conditions.
- Curated and annotated approximately **500–700 livestock images** in Roboflow and ran preliminary YOLO fine-tuning experiments, identifying the need for greater dataset scale and environmental diversity before field deployment.

Artificial Intelligence Intern

Mar 2024 – Dec 2024

CNIT – Master's Thesis Project, University of Pisa

Pisa, Italy

- Developed Python tooling for **resource-aware node-selection recommendations** and distributed PyTorch training on a **4-node heterogeneous bare-metal testbed**.
- Built a pre-training recommendation workflow using **SSH**, **CPU/memory checks**, and **iPerf3 pairwise throughput measurements** to expose compute availability and network conditions before launching PyTorch DDP jobs.
- Evaluated **PyTorch DDP**, **torchrun**, **Gloo**, **Docker**, **Docker Swarm**, **Wireshark**, and controlled network congestion using a **Spirent N4U traffic generator** with up to **90% network occupancy**.
- Under congestion, selecting a suitable two-node combination reduced one-epoch training time from **1,800 s to 820 s** (approximately **54% lower**) compared with using all four nodes; a poor two-node path required **2,716 s**.

Technical Skills

Generative AI & LLM Systems: Local LLMs, RAG, Conversational AI, Semantic Search, Embeddings, Vector Search, Context & Memory Management, LLM Evaluation, Prompt Engineering, Streaming Inference, Qwen, Ollama, FAISS

Machine Learning & Computer Vision: Python, PyTorch, Deep Learning, YOLO, OpenCV, Roboflow, Object Detection, Semantic Segmentation, Model Evaluation, Distributed Training

AI Systems & Backend: FastAPI, REST APIs, Streaming APIs, Model Serving, GPU Inference, Performance Benchmarking, Latency Profiling, Concurrent Inference, Docker, Linux, Git/GitHub

Edge AI & Embedded Deployment: Hailo-15H, NVIDIA Jetson, ONNX, Hailo DFC, INT8 Quantization, Yocto Linux, GStreamer, Real-Time Inference, Video Pipelines

Distributed Systems & Networking: PyTorch DDP, torchrun, Gloo, Docker Swarm, iPerf3, Wireshark, SSH, Resource Allocation, Network Performance

Cloud & Hardware: AWS, NVIDIA A Series, Jetson Nano, NVIDIA RTX Series, Hailo-15H

Education

M.Sc. Computer Science and Networking, University of Pisa	2021 – 2024
B.Eng. Computer Engineering, IAUT	2015 – 2019

Certifications

Google Cloud Fundamentals: Core Infrastructure | JNCIA-JUNOS | Hands-On AI: Computer Vision Projects with Ultralytics and OpenCV | Second Edition of the Advanced Hackathon on Machine Learning

Languages

English (C1 / Full Professional) | Italian (B1 / Intermediate proficiency) | Turkish (Native) | Persian (Native)